

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer with production RAG and agentic LLM systems for healthcare enterprise - bringing that end-to-end delivery discipline to ElevenLabs' most strategic voice AI deployments. Master's degree (Kent State University). Strong track record of measurable gains: ~35% retrieval precision improvement, >30% hallucination reduction, and ~30% fewer post-deployment defects. Proven ability to ship multi-provider AI integrations (Claude Sonnet, GPT-4o mini, Ollama) in production, own end-to-end delivery from data to API, and communicate AI system capabilities to both technical and non-technical stakeholders.

CORE COMPETENCIES

LLM Application Development • Retrieval-Augmented Generation (RAG) • Agentic Workflows • Multi-Provider AI Integration • FastAPI / Flask • Docker • Python • PyTorch • Hugging Face • Prompt Engineering • Evaluation Pipelines • End-to-End Delivery • Customer-Facing Technical Communication

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by **15-20%** on high-risk categories while holding precision >**90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ recorded sessions; reduced manual review time by **60-70%** (hours → <5 minutes per video).
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >**85%**.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped · Chrome Web Store

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-provider payload handling, seven-day translation cache, per-domain selector configs for 29 sites. Demonstrates production-grade multi-provider AI integration discipline transferable to any AI API modality.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; explainable risk scoring with audit-ready reasoning traces.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis. Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal APIs (Perplexity Sonar, Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%** over naïve direct-prompt orchestration.

EDUCATION

Master's Degree - *Kent State University, Ohio, USA*

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, Stable Diffusion, video analytics

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required